

PUSHKAR CHATURVEDI

Agentic AI Engineer — Machine Learning Engineer — Full-Stack AI Systems

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Education

Jaypee University of Engineering and Technology

Aug 2023 – Present

B.Tech in Computer Science and Engineering

CGPA: 9.4 / 10

Coursework: Probability & Statistics, Machine Learning, Linear Algebra, Data Structures & Algorithms, DBMS

Certifications: Andrew Ng ML Specialization (Coursera) | Mathematics for Machine Learning and Data Science (Coursera)

Currently Learning: AWS (Cloud Practitioner track)

Experience

Data & Backend Engineering Intern

May 2025 – Jul 2025

Inventlix (Remote) – Noida, India

- Built RESTful CRUD APIs with pagination, filtering, and sorting for a full-stack inventory management system using Django and React.
- Implemented secure JWT and Google OAuth authentication with role-based access control (RBAC), embedding user roles in token payloads to eliminate repeated database authorization lookups.
- Integrated Google Cloud Storage for image uploads and SendGrid for transactional emails (welcome, password reset, warranty alerts) with scheduled cron jobs for expiration notifications.

Projects

RepoSage — Agentic RAG Assistant for Codebase Understanding 🐙

Ongoing

Python, LangChain, FastAPI, Vector DB (Chroma), Celery, Docker

- Building a Retrieval-Augmented Generation pipeline that ingests GitHub repositories, chunking and embedding code and documentation into a vector database for context-grounded Q&A.
- Designing a LangChain agent with callable tools (codebase search, file retrieval, issue lookup) and conversation memory to help developers navigate unfamiliar repos and find open-source contribution opportunities.
- Implementing async repository ingestion via Celery and Redis, containerizing the service with Docker, and setting up GitHub Actions for CI.

EmotionSense — Multimodal Emotion-Aware AI Assistant 🐙

Ongoing

React.js, FastAPI, PyTorch, Hugging Face Transformers, Soundfile, OpenAI API

- Developing a full-stack multimodal AI assistant that integrates custom text and speech emotion recognition models (PyTorch, Hugging Face Transformers) with an LLM-powered conversational interface, exposed via FastAPI REST APIs for real-time interaction.
- Built multimodal inference pipelines combining speech emotion recognition and NLP models for context-aware, emotion-adaptive conversational AI.
- Designed a modular backend architecture separating AI inference, API services, and frontend communication to support scalable deployment and future multimodal extensions.

Offline Hindi Voice Assistant 🐙

Feb 2026

Python, Vosk, Coqui TTS, Raspberry Pi

- Built an offline speech AI pipeline (ASR + NLP + TTS) for Hindi interaction with optimized low-latency inference on ARM devices; recognized in **top 6 teams** at Bharat AI-SOC hackathon.

Technical Skills

Languages:	Python, SQL, JavaScript, Java, C++, C
Agentic AI / LLM:	LangChain, Retrieval-Augmented Generation (RAG), Vector Databases (FAISS, Chroma), OpenAI API, Prompt Engineering
AI/ML:	PyTorch, Hugging Face Transformers, Scikit-Learn, Stable-Baselines3, LLMs, NLP, Computer Vision, Speech Processing, Reinforcement Learning
Frameworks:	FastAPI, Django, Flask, React.js, REST APIs, JWT, OAuth
Databases:	PostgreSQL, Redis, Google Cloud Storage
Tools:	Docker, Git, Linux, Celery, GitHub Actions (CI/CD), Pandas, NumPy, Matplotlib

Leadership & Achievements

- Secretary – ROS Division, **ROSPINOT**, JUET; organized robotics workshops and collaborative technical projects.
- Organizing Team – CodeSrijan Hackathon; coordinated sponsorship outreach and event logistics.
- Top 25 Finalist – Bharat AI SoC | GATE 2026 (CS) AIR 7514 (Score: 500) | 3rd Place – Code Conquerors Hackathon | 1st Place – University Coding Competition